

RoBERTa, ResNeXt and BiLSTM with self-attention: The ultimate trio for customer sentiment analysis

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ABSTRACT

Sentiment analysis of customer feedback is a pivotal component of Natural Language Processing (NLP), enabling businesses to gauge consumer emotions towards their products and services. The inherent variability of language introduces substantial challenges in the analysis of unstructured customer data. Sentiment analysis techniques generally fall into two categories: traditional methods, which involve the extraction of hand-crafted features and the application of machine learning algorithms, and end-to-end deep learning approaches that process raw data, transforming them layer by layer into high-level textual features. Each method presents its own balance between simplicity and speed versus analytical power and flexibility, and they all grapple with the need for extensive labeled data and computational resources. To address these challenges, this paper presents an innovative hybrid model that amalgamates RoBERTa, ResNeXt, BiLSTM, and self-attention mechanisms. This integration capitalizes on the collective strengths of these models to surmount the limitations inherent in each. The proposed model proves to be a powerful and efficient tool for sentiment analysis, demonstrating proficiency across various data types and tasks. We undertake a comprehensive evaluation of the proposed model's accuracy and training efficiency using four benchmark datasets. Our investigation also explores the impact of different similarity measures and convolutional neural network architectures on the model's performance. The results affirm that our model not only achieves high accuracy but also significantly reduces training time compared to RoBERTa's fine-tuning. Furthermore, the model exhibits exceptional domain adaptability, particularly when fine-tuned on the IMDb dataset following initial training on the Yelp Review Full dataset. The practicality of the proposed model is underscored by its reduced computational complexity and its adeptness at navigating semantic and syntactic nuances.

1. Introduction

Understanding how consumers perceive the products or services they purchase online is a question of pivotal importance. This query propels businesses and researchers to probe into customer sentiment and feedback, elements that significantly influence strategic decision-making. The sentiment analysis market, which was valued at USD 3.15 billion in 2021, is projected to grow at a compound annual growth rate (CAGR) of 14.4% from 2022 to 2030, according to Polaris Market Research [1]. Sentiment analysis serves as a tool for enhancing customer satisfaction, loyalty, retention, and acquisition by providing profound insights into feedback, reviews, ratings, and preferences. Furthermore, it aids researchers in navigating through social media,

news articles, blogs, and other digital content to discern public opinion trends, sentiments, and emotional responses to a wide array of topics and events.

Sentiment analysis, a crucial task within the realm of natural language processing (NLP), strives to identify and categorize emotions or opinions embedded within texts. This spans from broad document-level analysis to the finer nuances of sentences, aspects, or entities [2]. This field grapples with numerous challenges, such as interpreting noisy and informal language, decoding negation and sarcasm, understanding context and domain-specific knowledge, processing multilingual and cross-lingual content, and maintaining fairness and ethical standards [3]. To address these complexities, there is a demand for robust

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Abbreviations			
Abbreviations	Definition	Abbreviations	Definition
BERT	Bidirectional Encoder Representations from Transformers	MLP	Multilayer Perceptron
BiLSTM	Bidirectional Long Short-Term Memory	NLP	Natural Language Processing
Caps	Capsule network	ReLU	Rectified Linear Unit
CBOW	Continuous Bag of Words	ResNet	Residual Network
CNN	Convolutional Neural Network	ResNeXt	Residual Networks with next dimension
CUDA	Compute Unified Device Architecture	RNN	Recurrent Neural Network
De-BERTa	Decoding-enhanced BERT with Disentangled Attention	RoBERTa	Robustly Optimized BERT Approach
ED	Euclidean distance	RMSProp	Root Mean Squared Propagation
ELMo	Embeddings from Language Models	SGD	Stochastic Gradient Descent
GLoVE	Global Vectors for Word Representation	SMOTE	Synthetic Minority Oversampling Technique
GPU	Graphics Processing Unit	SVM	Support Vector Machine
GRU	Gated Recurrent Unit	TS-SS	Triangle Area Similarity and Sector Area Similarity
IMDb	Internet Movie Database	UDA	Unsupervised Data Augmentation
LSTM	Long Short-Term Memory	VDCNN	Very Deep Convolutional Neural Network
MD	magnitude difference	Yelp-F	Yelp Review Full
MLM	Masked Language Model	Yelp-P	Yelp Polarity

and reliable models that are adept at navigating the intricate and varied landscape of natural language.

Transfer learning emerges as a pivotal technique in sentiment analysis, enabling models to assimilate knowledge from one domain and apply it to another. This adaptability is crucial for models to handle diverse datasets and tasks. Both classical and quantum machine learning methods can benefit from transfer learning. Classical approaches are known for their reliability and interpretability, with a proven track record across various applications. On the frontier of innovation, quantum machine learning promises exponential computational speedup and the capacity to process large datasets, potentially revolutionizing sentiment analysis [4]. Quantum methods, utilizing quantum random access memory, can expedite computation by harnessing

quantum superposition [5]. They also exploit quantum phenomena: superposition, entanglement, and interference, to intricately model cognitive aspects of sentiment such as ambiguity and sarcasm [6]. The synergy of classical and quantum approaches [7] opens avenues to address data scarcity, domain adaptation, model generalization, and the promotion of fairness and ethics by mitigating biases. Our model employs transfer learning to utilize RoBERTa's pre-trained capabilities for capturing semantic and syntactic nuances. Complementing this, we integrate classical machine learning techniques like ResNeXt, BiLSTM, and self-attention to refine text feature processing and enhance sentiment analysis performance.

Recent advancements in transformer-based models, notably BERT [8], have propelled natural language processing and sentiment analysis forward, primarily through the use of self-attention mechanisms to capture the contextual nuances of words. Despite these strides, the field continues to grapple with challenges such as domain adaptation [9], computational costs [10], and the intricacies of fine-tuning. Our proposed model seeks to address these concerns by integrating the strengths of RoBERTa [11], ResNeXt [12], BiLSTM [13,14], and self-attention [15]. Each component has demonstrated significant potential in NLP independently, and their combined capabilities are poised to overcome the existing limitations of sentiment analysis models.

The proposed system excels in contextual word embeddings, feature extraction, sequential dependency recognition, and feature prioritization for sentiment classification. Each component within this framework has demonstrated potential in natural language processing, and their collective integration effectively addresses prevalent challenges in sentiment analysis. The strategic combination of these technologies not only achieves competitive accuracy, rivaling state-of-the-art models like RoBERTa, but also stands out in training efficiency. Our model significantly reduces training time without sacrificing performance, showcasing its ability to maintain high accuracy with fewer parameters. This balance of precision and efficiency fills a vital research gap in sentiment analysis, presenting a model that is not just competitive in performance metrics but also finely tuned for optimal computational resource management.

This study is guided by key research questions that explore the integration and efficacy of advanced NLP components in sentiment analysis:

- How can RoBERTa, ResNeXt, BiLSTM, and self-attention be effectively combined to enhance sentiment analysis?
- How does the proposed model's accuracy and training efficiency compare to state-of-the-art models across various datasets?
- What is the impact of different similarity measures and convolutional neural network architectures on the performance of our model?

The structure of this paper provides a comprehensive exploration of our research. Section 2 reviews existing sentiment analysis methods and models, setting the stage for the introduction of our proposed model and an in-depth explanation of its components in Section 3. Section 4 details the datasets used to evaluate our model, while Section 5 delves into the hyperparameter tuning process and its impact on model performance. In Section 6, we present and discuss the experimental results, offering a comparison with state-of-the-art methods. Finally, Section 7 concludes the paper and outlines potential future research directions.

2. Related works

Sentiment analysis is a complex task that involves discerning sentiments within text data as positive, negative, or neutral. This requires a nuanced understanding of the contextual subtleties of language. Deep learning has emerged as a powerful approach to this challenge, with various techniques each addressing different aspects of sentiment analysis. In this section, we critically examine the literature, focusing on character-level, ensemble-based, transformer-based, and hybrid models. We trace the evolution of these methodologies, noting how

each subsequent model addresses the limitations of its predecessors. The following comparative analysis details the principal contributions and unique features of each method, their limitations, and how our model integrates these insights to advance the field.

2.1. Character-level models

Character-level models analyze text at its most granular level, capturing linguistic nuances that word-level models may overlook. By operating at the individual character level, these models bypass the need for word embeddings or language-specific pre-processing, allowing them to process a wide array of input characters, including those with misspellings, abbreviations, and emojis. This approach, while thorough, often requires significant computational resources, especially for longer texts.

Zhang et al. [16] demonstrated that character-level CNNs could match or exceed the performance of traditional methods, achieving notable accuracies across various datasets. Their work highlighted the impact of input representations, preprocessing, and network architectures on model performance, with their model attaining accuracies of 62.05% and 95.64% on the Yelp review full and Yelp polarity datasets, respectively. However, their model did not fully capture the sequential nature of text, which is crucial for sentiment analysis, and faced scalability challenges due to computational demands and extensive data requirements.

Conneau et al. [17] advanced the field with the VDCNN model, utilizing deep convolutional layers and pooling operations, inspired by ResNet's shortcut connections, to extract complex textual features. VDCNN achieved state-of-the-art results on the Yelp polarity dataset, with a remarkable 95.7% accuracy, and a notable 64.3% accuracy on the Yelp full dataset, surpassing other character-level models such as Char-CNN and fastText. However, it struggled with scaling to larger datasets and was sensitive to data noise.

These limitations underscore the need for a model that can effectively capture sequential dependencies in text while maintaining computational efficiency. Motivated by this need, our proposed model integrates character-level precision with broader contextual analysis. It combines RoBERTa with ResNeXt to enhance feature extraction at both character and word levels, thereby establishing a robust framework that overcomes the limitations of previous character-level models.

2.2. Ensemble-based models

Ensemble-based models leverage the collective strengths of various machine learning algorithms to surpass the accuracy and robustness of single-model approaches. However, these models often encounter challenges related to complexity and computational demands, which can impede their scalability. Our work introduces an innovative composite approach that judiciously selects and combines models, striking a balance between high performance and manageable computational load.

Tan et al. [18] developed an ensemble hybrid model that synergizes RoBERTa with LSTM, BiLSTM, and GRU, utilizing GloVe embeddings for sentiment analysis. This model was rigorously evaluated on diverse datasets such as IMDb, Twitter US Airline Sentiment, and Sentiment140, and was benchmarked against individual models such as RoBERTa, LSTM, BiLSTM, and GRU. The study revealed that majority voting outperformed averaging ensemble, particularly achieving an impressive 94.9% accuracy on IMDb. Nonetheless, the model's computational demands posed a challenge to its scalability.

In a systematic review, Tiwari et al. [19] explored bagging and boosting-based ensemble techniques within social network sentiment analysis, noting their potential to improve classifier accuracy and stability. However, they also identified significant challenges such as computational complexity, overfitting, and diversity loss associated with these techniques.

The challenges associated with ensemble-based models highlight the need for a model that can harness the collective strengths of various machine learning algorithms without the associated computational overhead. Motivated by this need, our proposed hybrid model builds upon these findings, merging RoBERTa with ResNeXt and BiLSTM, and is further enhanced by self-attention mechanisms. This combination adeptly handles the complexities of natural language data, offering a scalable and efficient solution for sentiment analysis.

2.3. Transformer-based models

Transformer-based models have revolutionized sentiment analysis, offering a groundbreaking ability to capture long-range dependencies and contextual nuances within text. Despite their transformative impact, these models often come with significant computational overhead, which can pose a barrier to their widespread adoption.

Devlin et al. [8] introduced BERT, a model that has reshaped the landscape of natural language processing. BERT's innovative use of a masked language model and next sentence prediction has enabled it to learn from vast amounts of unlabeled text, setting new benchmarks on GLUE and SQuAD. It achieved an impressive 93.5% accuracy on the SST-2 dataset, surpassing previous models. However, BERT's size and computational demands of BERT present challenges for efficiency and scalability.

In response to these limitations, Sanh et al. [20] proposed DistilBERT, a distilled version of BERT that employs a knowledge distillation technique. This process transfers the knowledge from a large teacher model (BERT) to a smaller student model (DistilBERT) by minimizing the difference between their outputs. This effectively maintains much of the original model's performance while reducing its size and speeding up inference. DistilBERT achieves 97% of BERT's performance metrics, yet is 60% faster and 40% smaller. It secures 91.3% accuracy on the SST-2 dataset with only half of BERT's parameters and layers. Nevertheless, DistilBERT still relies on substantial training data and computational resources, and its reduced linguistic capabilities and dependence on the teacher model pose new challenges.

Yang et al. [21] developed XLNet, an advanced autoregressive model that incorporates Transformer-XL technology to capture long-term dependencies and the relative positions of tokens [22]. This extension of the transformer model uses segment recurrence and relative positional encoding, enhancing XLNet's ability to surpass BERT by using permutation language modeling. This innovative approach allows XLNet to predict tokens in a sequence arbitrarily, addressing the pretrain-finetune discrepancy and the independence assumption inherent in BERT. XLNet achieved state-of-the-art results on diverse NLP tasks, including the GLUE benchmark and the SQuAD v2.0 question answering task, with a 93.35% accuracy on the SST-2 dataset. However, XLNet's increased computational cost and interpretability deficit remain significant obstacles.

He et al. [23] introduced DeBERTa, refining BERT's architecture and pre-training objectives with a focus on disentangled attention and an enhanced mask decoder. The disentangled attention mechanism employs dual vectors to represent the content and position of each word, enabling the model to compute attention weights with a nuanced understanding of text structure. The enhanced mask decoder leverages a masked language modeling objective that effectively predicts masked tokens and discriminates them from the rest of the vocabulary. DeBERTa has outperformed BERT and other transformer-based models on various NLP tasks, though it faces challenges such as increased model complexity and substantial memory demands.

Transformer-based models have revolutionized sentiment analysis, but their computational overhead and interpretability challenges have motivated our research to develop a more efficient and interpretable model. Our proposed model integrates RoBERTa with ResNeXt for streamlined feature dimensionality and simplified text embedding. Coupled with BiLSTM and self-attention mechanisms, it enables efficient feature extraction and enhanced interpretability. This marks a significant advancement in sentiment analysis, overcoming the limitations of previous transformer-based models.

2.4. Hybrid models

Hybrid models in sentiment analysis amalgamate the strengths of various neural network architectures to capture a broad spectrum of linguistic features. While offering comprehensive analysis, traditional hybrid models can often be rigid and inefficient. Our proposed model introduces a dynamic hybrid architecture that adapts to data complexity, ensuring high accuracy and improved efficiency.

Behera et al. [24] developed the Co-LSTM model, which combines convolutional and recurrent neural networks for sentiment analysis on large-scale social data. The model's three-layer architecture, consisting of an embedding layer using the CBOW method, a convolutional layer for feature extraction, and an LSTM layer for temporal dependencies, demonstrated superior accuracy and F1-score on Twitter and IMDB datasets. However, it lacked integration with pre-trained language models like BERT or RoBERTa for more contextualized word representations and did not include attention mechanisms for focused weighting of output vectors.

Beakcheol Jang et al. [25] introduced a hybrid model that merges CNN and BiLSTM with an attention mechanism to enhance text classification accuracy. This model employs skip-gram word embedding to represent words as vectors based on their context, harnessing CNN for high-level feature extraction and BiLSTM for capturing long-term word sequence dependencies. With these techniques, the model achieved an accuracy of 91.41% on the IMDB dataset, outperforming other models such as Multi-Layer Perceptron (MLP), CNN, LSTM, and hybrid models without attention. Despite its success, the model did not leverage pre-trained language models for nuanced word representations or advanced convolutional networks like ResNet or ResNeXt for complex feature extraction.

Dong et al. [26] created caps-BiLSTM, a model that integrates capsule networks with BiLSTM, employing GloVe for word embedding to encapsulate the semantic information of words. This innovative approach includes a convolutional component for local feature extraction, a capsule module enhanced by a modified dynamic routing algorithm for global features and text hierarchy, and a BiLSTM component for sentiment polarity interpretation. The model not only showed superior accuracy and robustness on sentiment analysis datasets but also outperformed the best existing models. For instance, on the IMDB dataset, caps-BiLSTM achieved an accuracy of 91.96%, surpassing the best baseline by 1.16%. Despite these advancements, the model faced challenges with computational complexity, capsule vector interpretation, and architectural diversity and flexibility.

Cach et al. [27] introduced a hybrid approach that combines LSTM, CNN, and SVM architectures, utilizing both BERT and Word2Vec embeddings to capture high-quality word and sentence representations. Their research not only demonstrated the efficacy of this method across eight diverse datasets of tweets and reviews but also highlighted the superiority of BERT embeddings over Word2Vec in generating word vectors that captured the semantic and syntactic features of words. The hybrid models consistently outperformed single-model approaches, with the BERT-CNN-LSTM configuration notably achieving an impressive 93.4% accuracy on the IMDB dataset, surpassing Word2Vec-CNN-LSTM by 3.7%. However, the models did not incorporate convolutional networks like ResNet or ResNeXt for intricate feature extraction, nor did they include attention mechanisms such as self-attention for enhanced output vector weighting.

Zhou et al. [28] introduced KSCB, an unsupervised sentiment analysis model that addresses class-imbalance and the challenges of an unlabeled corpus. KSCB integrates K-means++, SMOTE, CNN, and BiLSTM to adjust data distribution and automate sentiment document labeling. Evaluated against five supervised methods on various datasets, KSCB showed commendable performance, achieving up to 90.8% accuracy on the IMDB dataset. The model utilizes GloVe embeddings for high-quality word and sentence representations. Despite its strengths, KSCB does not incorporate attention mechanisms like self-attention or capsule

networks for output vector weighting, nor does it employ Transformer models such as BERT or RoBERTa for more compact and meaningful word embeddings.

Tan et al. [29] crafted a hybrid deep learning model that adeptly combines the analytical strengths of Transformer and RNN models to address the intricacies of sentiment analysis. This model confronts the challenges posed by lexical diversity, imbalanced datasets, and long-distance dependencies within texts. Employing a data augmentation strategy alongside GloVe word embedding, the model juxtaposes RoBERTa with LSTM to achieve an accuracy of 92.96% on the IMDB dataset, 91.37% on the Twitter US Airline Sentiment dataset, and 89.70% on the Sentiment140 dataset. However, it does not integrate convolutional networks like ResNet or ResNeXt for extracting more complex text features, nor does it incorporate attention mechanisms for nuanced output vector weighting.

Kaur et al. [30] developed a hybrid feature extraction method for sentiment classification in text reviews, structured into three pivotal steps: pre-processing, review-related features, and aspect-related features. The pre-processing phase utilizes NLP techniques to refine and standardize the text input. The core of the method lies in its hybrid feature extraction, which amalgamates review-centric features that gauge the overall sentiment with aspect-centric features that delve into specific opinions and emotions related to various facets of a product or service. An LSTM classifier is then employed to categorize each review into positive, negative, or neutral sentiments. This method has shown to outshine single feature extraction approaches in performance. However, it does have its constraints, primarily its reliance on predefined aspects which may not fully encapsulate the intricate opinions and emotions of consumers, and its potential struggle with complex reviews that contain multiple or conflicting sentiments.

While hybrid models have made significant strides in sentiment analysis, they often grapple with computational complexity and struggle to fully capture intricate opinions and emotions in reviews. Some models are limited by their reliance on predefined aspects and lack advanced feature extraction capabilities. Motivated by these challenges, our research introduces a dynamic hybrid model that integrates RoBERTa, ResNeXt, BiLSTM, and self-attention mechanisms. This approach aims to address the limitations of previous hybrid models.

3. Methods

The task of customer sentiment analysis necessitates a multifaceted understanding of language, requiring the extraction of salient features, recognition of word dependencies, and evaluation of token significance for sentiment prediction. In light of the discussions in Section 2, we propose a hybrid model that integrates the strengths of RoBERTa, ResNeXt, BiLSTM, and self-attention to address the comprehensive challenges of this task (see Fig. 1). Our model is architecturally composed of four pivotal modules:

1. **Word Embedding Module:** This module utilizes RoBERTa to tokenize and encode input text into a matrix of word embeddings $\mathbf{X} \in \mathbf{R}^{n \times d}$, where n is the token count and d represents the embedding dimension. To enhance computational efficiency, we employ only the last four layers of RoBERTa, merged into a single layer with a unified attention head, resulting in an advanced matrix of word embeddings $\mathbf{X}' \in \mathbf{R}^{n \times d}$. This strategy yields a rich yet computationally efficient feature set.
2. **Feature Extraction Module:** This module processes the advanced word embeddings $\mathbf{X}'$ to distill essential features, culminating in a matrix $\mathbf{F} \in \mathbf{R}^{m \times k}$ with m representing the streamlined token count and k the augmented feature dimension. The ResNeXt framework, with its diverse convolutional groups and distinct cardinalities, adeptly accommodates variable-length input sequences through global average pooling, capturing a comprehensive array of textual features.

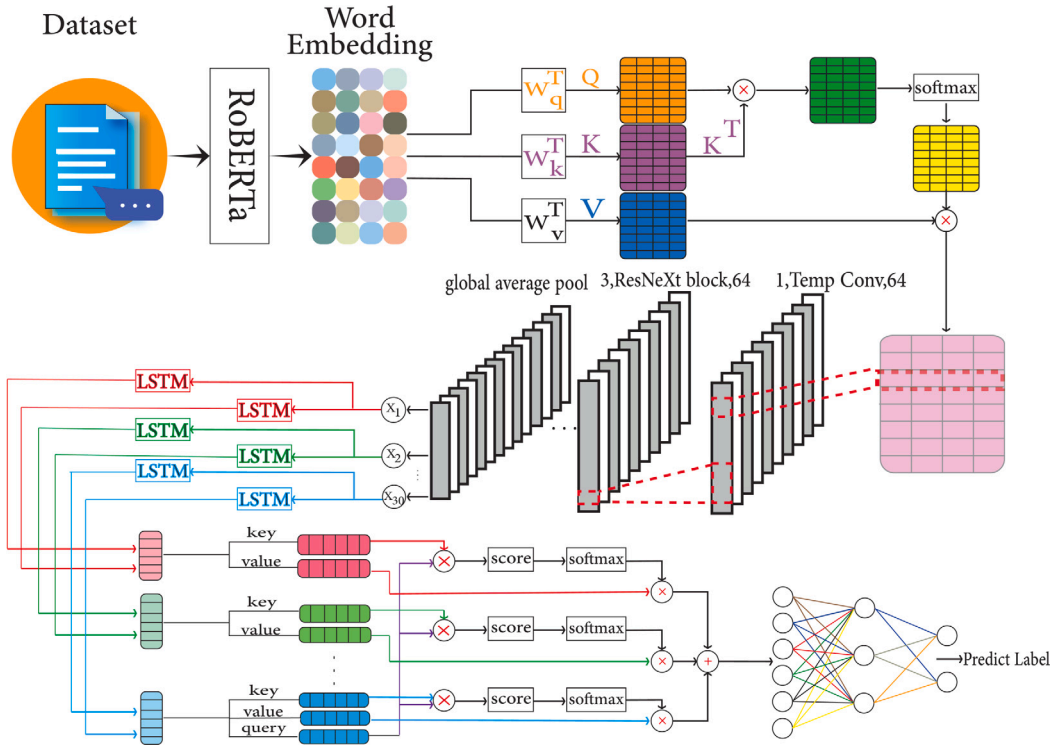


Fig. 1. The architecture of our hybrid model for sentiment analysis.

3. **Sequence Modeling Module:** The BiLSTM layer processes the feature vectors F capturing the sequential and contextual features of words. Operating bidirectionally, BiLSTM enhances text comprehension by integrating both antecedent and subsequent information into each word representation. The output is a matrix of word representations $H \in \mathbb{R}^{m \times h}$ where h is the hidden dimension, providing a deeper contextualization.
4. **Self-Attention Mechanism:** This mechanism assigns weights to the word representations H , calculating distinct weights for each token based on its relevance to the sentiment analysis task. Our model employs the TS-SS similarity measure [31], which accounts for both the temporal and spatial dimensions of the feature vectors. The result is a vector of weighted word representations $S \in \mathbb{R}^h$, representing the aggregate of the word representations scaled by their respective weights. This selective emphasis on pertinent features serves to enhance the model's performance and accuracy in sentiment analysis.
5. **Sentiment Classification Layer:** The culmination of our model's processing is the classification of the weighted word representations S into sentiment categories. Utilizing a softmax activation function, this fully connected layer outputs a probability vector $P \in \mathbb{R}^2$, which reflects the likelihood of each sentiment class. By learning the associations between input tokens and sentiment classes, the layer provides a confident prediction of the text's sentiment.

To operationalize the described architecture, we delineate the following algorithmic workflow, which succinctly encapsulates the sequential processing of input data through the various layers of our model, culminating in the sentiment classification:

Algorithm 1 Enhanced Hybrid Sentiment Analysis

- 1: **Input:** A collection of customer reviews with variable lengths
- 2: **Output:** Sentiment classification for each review
- 3: **Begin**
- 4: Preprocess the input text to ensure uniform length:

- 5: Truncate or pad sequences as necessary to fit the model's input size requirements.
- 6: **for** each review in the dataset **do**
- 7: // Word Embedding with RoBERTa
- 8: $X \leftarrow$ Generate contextual word embeddings for each token using RoBERTa.
- 9: // Attention Mechanism with a Unified Head and TS-SS
- 10: $X' \leftarrow$ Apply a unified attention head using TS-SS similarity measure to the last four layers of RoBERTa embeddings, obtaining an advanced matrix of word embeddings.
- 11: // Feature Extraction with ResNeXt
- 12: $A \leftarrow$ Process the advanced embeddings X' through ResNeXt blocks to extract pertinent features.
- 13: $F \leftarrow$ Apply global average pooling on A to obtain a uniform feature representation.
- 14: // Sequence Modeling with BiLSTM
- 15: $H \leftarrow$ Capture sequential and contextual dependencies within the text using a BiLSTM layer.
- 16: $Q \leftarrow$ Employ the last hidden state of the BiLSTM as a query vector in the attention mechanism.
- 17: // Self-Attention Mechanism
- 18: $S \leftarrow$ Integrate information from preceding hidden states using the attention mechanism with TS-SS scores.
- 19: // Sentiment Classification
- 20: $P \leftarrow$ Feed the attention mechanism's output into a dense layer to classify sentiment.
- 21: Output the sentiment class with the highest probability in P .
- 22: **end for**
- 23: **Return** the probability distribution over sentiment classes for each review.
- 24: **End**

The pseudocode above provides a high-level overview of the model's functionality, bridging the gap between the conceptual framework and practical implementation. It is intended to guide the reader through the logical sequence of operations that underpin our sentiment analysis approach.

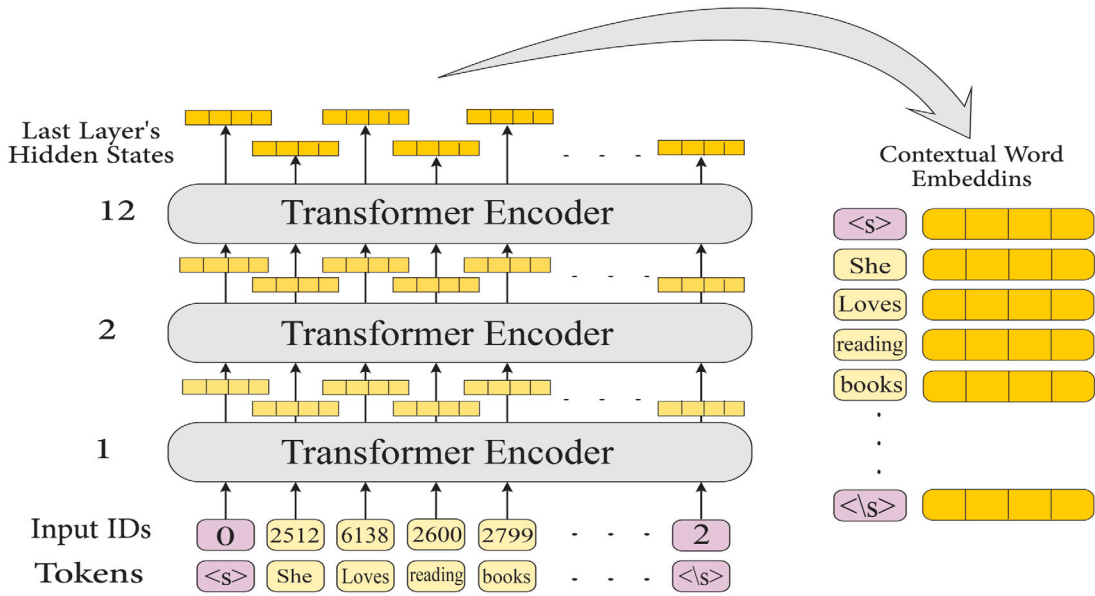


Fig. 2. An example of how RoBERTa encodes an input text into word embeddings.

The following subsections will provide detailed descriptions of each module within our model, elucidating the intricate processes and the rationale behind their design. This in-depth exploration will further substantiate the model's capabilities and its applicability to the domain of sentiment analysis.

3.1. RoBERTa: The word embedding powerhouse

RoBERTa Base serves as the foundational word embedding module in our hybrid model. This pre-trained language model, built upon transformer architectures, is renowned for its self-attention mechanisms that process sequential data in parallel, thereby enhancing the efficiency and performance of language modeling tasks [15]. RoBERTa Base has been trained on an extensive corpus of English data, including Book-Corpus and English Wikipedia, utilizing a masked language modeling (MLM) objective. This approach involves randomly masking words in the input text and prompting the model to predict them based on the contextual clues provided by the surrounding text (see Fig. 2). With its 12 layers, 12 attention heads, and a staggering 125 million parameters, RoBERTa Base is adept at capturing the semantic and syntactic features of words across various contexts and domains. However, it is important to note that RoBERTa Base is not inherently fine-tuned for any specific downstream task. For instance, to tailor it to sentiment analysis objectives, under normal circumstances, additional training data and labels would be used to adapt it to these specific tasks. In our methodology, we harness RoBERTa Base solely for its word embedding capabilities, deliberately eschewing the first token of its output, which is typically employed for text classification purposes. Instead, we focus on the rich word embeddings it generates, which are subsequently channeled into the feature extraction module of our model. Moreover, RoBERTa Base distinguishes itself from BERT by adopting a sequence length of 512 tokens, a threshold that BERT only utilized in 10% of instances. The distinctions between RoBERTa and BERT are pivotal to our model's design:

- **Pretraining Objective:** RoBERTa discards BERT's next-sentence pretraining objective, which was initially intended to aid in learning sentence-level representations but was ultimately deemed ineffective. RoBERTa relies exclusively on MLM for pretraining.

- **Training Parameters:** RoBERTa's training is conducted with substantially larger mini-batches and learning rates, facilitating faster convergence and superior outcomes. Specifically, RoBERTa employs 8k sequences and a learning rate of $5e-4$, in contrast to BERT's 256 sequences and a learning rate of $1e-4$.
- **Data Utilization:** RoBERTa is trained on 160 GB of text, encompassing a broader and more extensive range of texts, whereas BERT's training is limited to 16 GB of text.
- **Tokenization Technique:** RoBERTa utilizes a byte-level BPE tokenizer that segments words into sub-words based on character-level patterns, enhancing its ability to manage rare and out-of-vocabulary words more effectively than BERT's word piece tokenizer, which relies on frequency-based splits.

Distinct from other word embedding methods such as Word2Vec, FastText, and ELMo, RoBERTa utilizes a broader context than Word2Vec's models, which are based on a shallow neural network that learns word associations within a limited context window [32]. It overcomes the limitations of Word2Vec, such as the inability to encode unknown or out-of-vocabulary words due to its fixed vocabulary. Word2Vec also does not consider word order, which can lead to semantic ambiguities and a loss of syntactic and semantic understanding in sentences with similar word composition but different meanings. For example, the sentences "She likes him more than me." and "She likes me more than him." would be indistinguishable in Word2Vec's vector space. In contrast, RoBERTa's transformer architecture and MLM objective allow it to grasp the syntactic and semantic subtleties of sentences.

FastText enhances Word2Vec by integrating sub-word elements, allowing for effective management of rare and out-of-vocabulary terms [33]. For instance, without 'commonly' in its training data, FastText can deduce its embedding by aggregating the embeddings of its sub-components, such as <co>, com, omm, mmo, mon, etc. This approach resolves the issue of words missing from the training corpus. Nonetheless, FastText, akin to Word2Vec, overlooks word order, which may affect syntactic and semantic comprehension. In contrast, RoBERTa utilizes its transformer structure and MLM objective to surmount this hurdle, providing a richer understanding of sentence context and a more intricate representation of language.

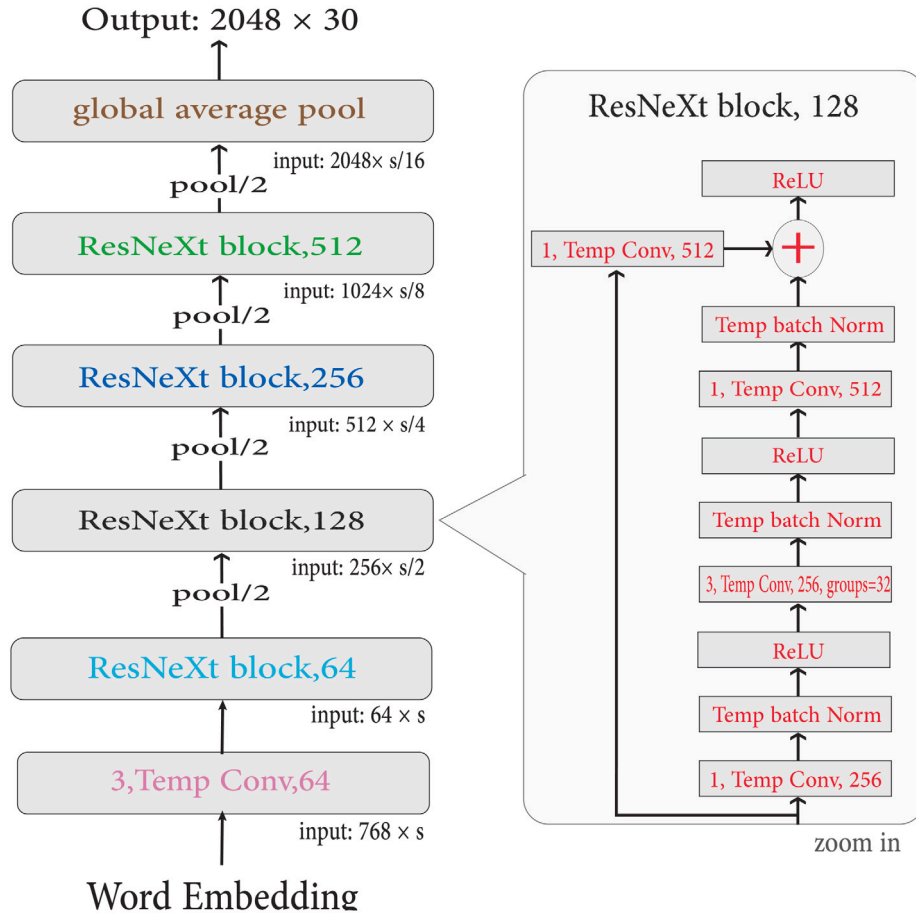


Fig. 3. An example of a ResNeXt block with 32 branches.

ELMo employs bidirectional LSTM networks to create contextualized word embeddings, capturing both syntactic and semantic information across different contexts [34]. Despite its strengths, ELMo falls short in several areas where RoBERTa excels:

- **Parallel Processing:** ELMo's sequential processing is slower compared to RoBERTa's parallel processing capabilities.
- **Self-Attention Mechanisms:** ELMo lacks self-attention mechanisms, which are crucial for capturing long-range dependencies between words, a feature inherent in RoBERTa's design.
- **Pretraining Objective:** ELMo does not utilize an MLM objective for pretraining, which is instrumental in learning effective bidirectional representations, a core aspect of RoBERTa's training process.

In our model, we do not fine-tune RoBERTa, opting instead to leverage its pre-trained strengths to enrich the feature extraction process. This strategic choice allows us to maintain computational efficiency while benefiting from RoBERTa's deep understanding of language nuances.

3.2. ResNeXt: Enhancing feature extraction

ResNeXt, introduced by Xie et al. [12], represents a significant advancement in deep neural network architecture for image classification. The term "ResNeXt" stands for Residual Networks with an added dimension, cardinality, which refers to the number of parallel paths or branches within each residual block. This innovative architecture posits that increasing the cardinality of a network can more effectively boost its performance than simply deepening or widening it. ResNeXt offers several key advantages over its predecessor, ResNet, another

deep neural network architecture renowned for its residual connections that facilitate the training of very deep networks without compromising training accuracy [35]. ResNeXt's design is both simpler and more modular compared to ResNet, leading to fewer hyper-parameters and easier scalability. Moreover, ResNeXt's higher representational capacity allows it to learn a more complex and diverse set of features from data. In various image classification benchmarks, including ImageNet-1K, ImageNet-5K, and COCO detection, ResNeXt has consistently outperformed ResNet. Fig. 3, shows an example of a ResNeXt block with 32 branches.

A typical ResNeXt block comprises a series of parallel convolutional layers sharing the same topology, followed by a 1×1 convolution that amalgamates their outputs. These parallel layers, or branches, each with a low-dimensional embedding and a transformation function (such as a convolution or a fully connected layer), process the input feature map in distinct ways. The block's output is then combined with the input via a residual connection, enhancing gradient flow and mitigating the vanishing gradient problem. The ResNeXt block's formulation is given by:

$$F(x) = x + \sum_{i=1}^C T_i(x)$$

where $F(x)$ denotes the block's output, C represents the cardinality, and $T_i(x)$ is the transformation function for the i th branch.

In the proposed model, we adapt the ResNeXt architecture for text classification at the sub-word level. We employ 9 layers of ResNeXt blocks, each with a cardinality of 32, meaning every block contains 32 parallel convolutional branches with identical topologies. Following the ResNeXt network, we apply a global average pooling operation to produce 30 tokens for each sentence, averaging the tokens within

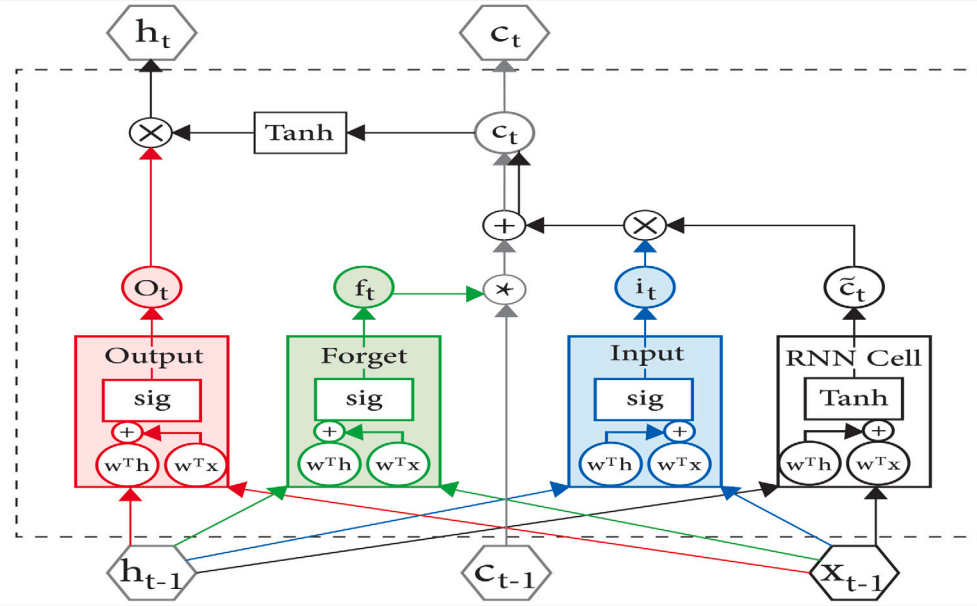


Fig. 4. The architecture of an LSTM layer.

each kernel. Each token is 512-dimensional, serving as the input for the subsequent stage of the introduced model.

This approach allows our model to extract salient and diverse features from text, providing a robust foundation for the sequential and contextual processing that follows in later stages of the sentiment analysis task.

3.3. LSTM: Long-term dependency mastery

Long Short-Term Memory (LSTM) networks are a sophisticated variant of Recurrent Neural Networks (RNNs) capable of learning long-term dependencies in sequential data, a feat that standard RNNs struggle with due to the vanishing or exploding gradient problem [36]. LSTMs are equipped with a unique gating mechanism consisting of three gates: the input gate, forget gate, and output gate. These gates effectively regulate the flow of information, allowing the network to retain or discard information as needed. Fig. 4, shows the architecture of an LSTM layer that processes one token of an input sequence at a time. The architecture of an LSTM layer is designed to process one token of an input sequence at a time, with the following equations governing the gates' operations:

- Input Gate:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i)$$

- Forget Gate:

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$$

- Output Gate:

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$$

- Cell Input Activation Vector:

$$\tilde{c}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c)$$

- Cell State Vector:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

- Hidden State Vector:

$$h_t = o_t \odot \tanh(c_t)$$

where, σ denotes the sigmoid activation function, and $\odot$ represents element-wise multiplication. The weight matrices W and bias vectors b are parameters learned during training.

LSTM networks offer several improvements over conventional RNNs, including:

- **Long-Range Dependencies:** LSTMs can capture dependencies from distant past, enabling them to utilize historical information for current or future predictions.
- **Gating Mechanisms:** These mechanisms help LSTMs prevent overfitting and enhance generalization by focusing on relevant features and discarding noise.

3.4. BiLSTM: Bidirectional contextual insights

Bidirectional LSTM (BiLSTM) is an extension of LSTM that processes sequential data in both forward and backward directions, granting it access to both past and future context. This bidirectional processing is crucial for tasks like natural language processing and sentiment analysis, where understanding the full context of text is essential.

A BiLSTM consists of two LSTM layers that read the input sequence from opposite ends. The outputs of these layers are concatenated at each time step to form the final output vector:

$$y_t = [h_t^{\rightarrow}; h_t^{\leftarrow}]$$

where y_t is the final output vector; $h_t^{\rightarrow}$ and $h_t^{\leftarrow}$ are the forward and backward hidden state vectors respectively; $[\cdot]$ denotes concatenation. BiLSTM networks, enhancing LSTM, offer benefits including:

- **Contextual Understanding:** By capturing both past and future contexts, BiLSTMs can discern the sentiment of words within varying contexts, such as distinguishing the sentiment of "great" in different parts of a sentence.
- **Contextual Understanding:** With two distinct LSTM layers, BiLSTMs can learn a more diverse set of features, enhancing the performance and accuracy of sentiment analysis.

3.5. Self-attention: Focusing on what matters

Self-attention is a transformative technique in the field of natural language processing, enabling models to discern intricate relationships

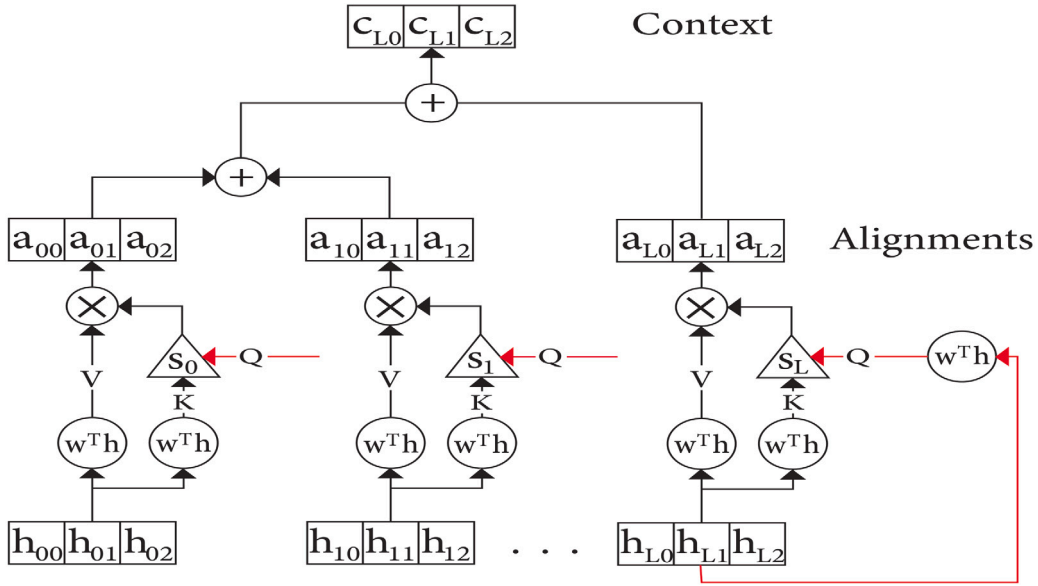


Fig. 5. An example of self-attention for sentence “She loves reading books”.

within a sequence of elements, such as words in a sentence. This mechanism is particularly advantageous for tasks that require a nuanced understanding of language, as it can highlight critical features while filtering out irrelevant noise. Essentially, self-attention allows a model to focus on the most pertinent aspects of the data for a given task.

The power of self-attention lies in its use of three distinct vectors: query, key, and value, derived from the input sequence through various transformations. The query vector represents the element under scrutiny, the key vector corresponds to another element for comparison, and the value vector contains the information to be extracted. The interaction between these vectors is pivotal in determining the relevance of different parts of the sequence:

- To ascertain the significance of a key for a given query, their vectors are multiplied, and a softmax function is applied to the result. This produces a weight that quantifies the importance of the key to the query.
- The output for a query is computed by weighting each value vector according to these calculated weights and summing them, yielding a new vector rich in information pertinent to the query.

The mathematical formulation of self-attention is expressed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where, Q is the matrix of query vectors, k is the matrix of key vectors, V is the matrix of value vectors, and d_k is the dimensionality of each key vector. The Fig. 5, illustrates how self-attention works.

Consider the sentence “She loves reading books.” To focus on the word “books”, the following steps are taken:

1. Each word is transformed into query, key, and value vectors, encoding the word in different representations.
2. The query vector for “books” is compared with the key vectors for all words, generating a score for each pair that reflects the relationship between “books” and the other words. These scores are then converted into weights using a softmax function.
3. The value vectors are combined using these weights, creating a new vector for “books” that encapsulates the most relevant information based on its connections to the other words in the sentence.

This self-attention mechanism enables our model to effectively prioritize and synthesize information, enhancing its ability to perform language-related tasks with greater precision and insight.

3.6. TS-SS: A geometric approach to document similarity

TS-SS is an innovative geometric method we employ within the self-attention layer to measure document similarity, offering a novel alternative to the standard dot product approach. By considering multiple perspectives, TS-SS calculates similarity through the multiplication of two components: triangle area similarity (TS) and sector area similarity (SS). TS represents the area of the triangle formed by two vectors and the origin, while SS denotes the area of the sector formed by the same vectors and the arc between them. The TS-SS formula is:

$$TS - SS(\vec{u}, \vec{v}) = TS(\vec{u}, \vec{v}) \times SS(\vec{u}, \vec{v})$$

where:

- Triangle Area Similarity:

$$TS(\vec{u}, \vec{v}) = \frac{1}{2} \|\vec{u}\| \|\vec{v}\| \sin \theta'$$

- Adjusted Angle:

$$\theta' = \arccos(\cosine(\vec{u}, \vec{v})) + 10$$

- Sector Area Similarity:

$$SS(\vec{u}, \vec{v}) = \pi \cdot (ED(\vec{u}, \vec{v}) + MD(\vec{u}, \vec{v}))^2 \cdot \frac{\theta'}{360}$$

In these equations, θ' is the angle between vectors $\vec{u}$ and $\vec{v}$, and $|\cdot|$ is the Euclidean norm, $ED(\vec{u}, \vec{v})$ is the Euclidean distance, and $MD(\vec{u}, \vec{v})$ is the magnitude difference. TS-SS values range from 0 (perfect similarity) to infinity (no similarity), capturing the orientation, proximity, and magnitude differences between vectors. This comprehensive approach allows TS-SS to detect subtleties in document similarity with greater finesse than traditional methods, leading to state-of-the-art results in unsupervised text clustering.

3.7. Dense layer: The synthesis of learning

In our RoBERTa-ResNeXt-BiLSTM model, the dense layer, or fully connected layer, plays a crucial role in synthesizing the learned features. Each dense layer connects every input from the preceding layer

to every activation unit it contains, ensuring a thorough integration of learned patterns. Our model incorporates two dense layers:

1. **Hidden Layer:** With 512 neurons, this layer is responsible for extracting patterns and features that correlate the input tokens with the output classes.
2. **Classification Layer:** This layer, equipped with one neuron per class in the dataset, employs the softmax activation function to calculate the probability distribution across the classes for the input tokens.

3.8. Complexity analysis: Assessing efficiency and scalability

In the domain of algorithmic design and computational analysis, a thorough understanding of time and space complexity is of paramount importance. It offers critical insights into the efficiency and scalability of an algorithm, both of which are vital considerations in practical applications. As we delve deeper into the intricacies of our hybrid model, we will conduct a meticulous examination of the time and space complexity associated with each integral component: RoBERTa, ResNeXt, BiLSTM, Self-Attention, and Dense Layer. This analytical exploration will shed light on how these components collectively contribute to the model's performance in sentiment analysis tasks.

3.8.1. RoBERTa

The RoBERTa component comprises multiple layers. Each of these layers incorporates a self-attention mechanism, layer normalization, and a feed-forward neural network. The cumulative time complexity for each layer in RoBERTa is represented as:

$$O(H \times (n \times d(n + d + m + 1)))$$

where H represents the number of heads in the attention mechanism, n denotes the number of tokens, d signifies the dimension of each token, and m corresponds to the size of the first layer in the feed-forward network. This complexity originates from the computation involved in the self-attention mechanism ($O(n^2 \times d + d^2 \times n)$), layer normalization ($O(n \times d)$), and the feed-forward network ($O(n \times (d \times m + m \times d))$). In terms of space complexity, given that we utilize a pre-trained RoBERTa model without any fine-tuning, the total space complexity is:

$$O(n \times d)$$

this complexity arises from the storage requirements for the outputs corresponding to each token.

3.8.2. ResNeXt

The ResNeXt component, designed to extract essential features from word embeddings, is composed of multiple blocks. Each block integrates a convolutional operation, batch normalization, and a residual connection. The overall time complexity of each block in ResNeXt is:

$$O(C \times (N \times W \times (D + W + F' + 1) + N \times F' \times (D + 1)))$$

where C is the cardinality, N is the number of input features (in this case, tokens), W is the number of filters in the first convolutional layer, D is the dimension of the input (in this case, the embedding dimension), and F' is the number of filters in the third convolutional layer. The complexity of each block in the ResNeXt component, which is responsible for extracting essential features from word embeddings, arises from several operations. The convolutional operation, which includes $O(N \times D \times W)$ for the first convolutional layer, $O(N \times W^2)$ for the second convolutional layer, and $O(N \times W \times F')$ for the third convolutional layer, contributes significantly to the complexity. Additionally, the batch normalization operation, which includes $O(N \times W)$ after the first and second convolutional layers and $O(N \times F')$ after the third convolutional layer, also adds to the complexity. Lastly, the residual connection, represented by $O(N \times D \times F' + N \times F')$, is another contributing factor to the overall complexity.

Regarding the space complexity, the ResNeXt model necessitates storage for the weights and parameters associated with each block. The space complexity for these weights and parameters is represented as:

$$O(C \times (W \times (D + W + M + M') + F' \times (W + D + M'')))$$

in this representation, M denotes the spatial dimensions of the feature map of the first convolutional layer, M' signifies the spatial dimensions of the feature map of the second convolutional layer, and M'' corresponds to the spatial dimensions of the feature map of the third convolutional layer and the residual connection convolutional layer output.

3.8.3. BiLSTM

The BiLSTM component, proficient in capturing sequential and contextual features of words, has a time complexity that is approximately twice that of LSTM. This is attributed to the presence of two distinct LSTM layers in its architecture. Similarly, the space complexity of BiLSTM is also approximately double that of LSTM. The LSTM component exhibits a time complexity of:

$$O(T \times h \times (d + h + 1))$$

where T represents the number of time steps, h denotes the size of the hidden state, and d signifies the size of the input dimension. Each gate within LSTM has a time complexity of $O(h \times (d + h + 1))$, and each element-wise operation exhibits a time complexity of $O(h)$. The space complexity of LSTM is:

$$O(h \times (d + h + T + 1))$$

which arises from the storage requirements for weights and parameters for each gate $O(d \times h + h^2 + h)$, the hidden states and cell states $O(T \times h)$, and the intermediate computations $O(T \times h)$.

3.8.4. Self-attention

The Self-Attention component plays a pivotal role in assigning weights to word representations. This process is streamlined by using only the last hidden state of BiLSTM as a query, thereby focusing on the most relevant features for sentiment analysis. The time complexity of this Self-Attention component is:

$$O(n \times d + n \times d^2)$$

where, n denotes the sequence length, and d signifies the dimensionality of the input. The computation of the attention scores contributes a complexity of $O(n \times d)$, and the linear transformations contribute a complexity of $O(n \times d^2)$. The space complexity of this Self-Attention component is:

$$O(d^2 + n \times d + n)$$

this complexity arises from the storage requirements for the weights $O(d^2)$, the attention scores $O(n)$, and the intermediate variables $O(n \times d)$.

3.8.5. Dense layer

The Dense Layer component plays a crucial role in our model by classifying the weighted word representations into sentiment categories. The time complexity of the Dense Layer is:

$$O(d \times m + m \times c)$$

Here, d is the dimensionality of the input vector, m is the dimensionality of the output from the first linear transformation, and c is the number of classes. The complexity components stem from two operations: the initial linear transformation, which exhibits a time complexity of $O(d \times m)$, and the subsequent linear transformation, which has a time complexity of $O(m \times c)$. The space complexity of the Dense Layer is:

$$O(d \times m + m \times c + c)$$

this complexity arises from the storage requirements for the weights and outputs of each linear transformation.

4. Dataset

In our research, we utilize four diverse datasets for sentiment analysis, each offering a unique perspective on sentiment classification across different contexts and complexities. The datasets are as follows:

The IMDb Dataset includes 50,000 movie reviews from the Internet Movie Database, evenly divided into 25,000 training and 25,000 testing samples [37]. Reviews are binary-labeled as positive or negative, with each review presented as a raw text string. The maximum sequence length is set at 512, making the IMDb dataset a standard benchmark for binary sentiment classification.

The TweetEval (Sentiment) Dataset is a part of the larger TweetEval benchmark, which is a collection of seven diverse and challenging tasks in Twitter, all framed as multi-class tweet classification [38]. This particular dataset focuses on the sentiment task, which involves classifying tweets into three categories: positive, neutral, or negative, based on the emotion or attitude conveyed by the tweet authors. It comprises 45,389 training, 2000 validation, and 11,906 testing samples, with each tweet presented as a raw text string. The maximum sequence length for this dataset is 100, providing a valuable resource for sentiment analysis on social media data.

The Yelp-Polarity Dataset is derived from the broader Yelp reviews dataset, which encompasses reviews from 11 metropolitan areas across the USA and Canada [16]. This dataset provides insights into 150,346 businesses across diverse categories, including restaurants, shopping, beauty, and nightlife. In terms of sentiment analysis, reviews with 1 and 2 stars are categorized as negative, while those with 3 and 4 stars are considered positive. This classification excludes 5-star reviews to minimize noise. The dataset is substantial, comprising 560,000 training samples and 38,000 testing samples. Each review is presented as a raw text string, and the maximum sequence length is capped at 512. The Yelp-Polarity Dataset serves as a comprehensive resource for binary sentiment classification tasks.

The Yelp-Review-Full Dataset, a subset of the Yelp reviews dataset, maintains the original 1 to 5 star ratings as labels for the reviews, randomly selecting 130,000 training and 10,000 testing samples for each star rating [16]. With a total of 650,000 training and 50,000 testing samples, each review is provided as a raw text string. The Yelp-Review-Full dataset is used for fine-grained multi-class sentiment classification and has a maximum sequence length of 512.

Collectively, these datasets provide a robust platform for assessing the performance of sentiment analysis models, catering to a wide range of data types and classification complexities. They include both binary and multi-class sentiment classification tasks, sourced from online platforms and social media, ensuring a comprehensive evaluation of the model's capabilities.

5. Hyperparameter optimization strategy

The hyperparameter tuning process is a critical step in refining the performance of our RoBERTa-ResNeXt-BiLSTM model. This procedure involves the meticulous evaluation of three key hyperparameters: the learning rate, the number of LSTM units, and the optimizer. To achieve this, we employ a comprehensive grid search method that exhaustively tests all permutations of these hyperparameters across four distinct datasets. The objective is to pinpoint the combination that delivers the most accurate validation results, as gauged by the model's ability to correctly predict labels for reviews. Central to this process is the cross-entropy loss function, which serves as a measure of the model's predictive accuracy for multi-label classification tasks. It is mathematically represented as:

$$L(Y, \hat{Y}) = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log \left(\frac{e^{z_{ij}}}{\sum_{k=1}^C e^{z_{ik}}} \right)$$

where, Y denotes the matrix of actual labels, and $\hat{Y}$ signifies the matrix of predicted probabilities, derived from applying the softmax function

to the input scores z_{ij} . The variables N and C correspond to the number of samples and classes, respectively.

The learning rate is a pivotal factor that influences the extent of weight adjustments during the gradient descent steps. To optimize this parameter, we experiment with three distinct values: $1e-3$, $1e-4$, and $1e-5$. Additionally, we implement a linear learning rate schedule complemented by a warmup phase. This approach begins with a low learning rate that incrementally ascends to the initial value before methodically descending to zero as training progresses. Such a strategy is designed to mitigate abrupt gradient updates at the onset of training, thereby fostering stable convergence.

Furthermore, the number of LSTM units is instrumental in determining the capacity of the hidden state vector to encapsulate sequential information within the input text. We evaluate three configurations, comprising 256, 512, and 768 LSTM units, to ascertain the optimal size. While an increased number of LSTM units can potentially amplify the model's expressive power and its proficiency in learning long-range dependencies, it also escalates the risk of overfitting and elevates computational complexity.

In the pursuit of refining our model, the optimizer emerges as a crucial component that updates the model weights in response to the gradients of the loss function. We explored three distinct optimizers, each with its unique approach to managing the learning rate and addressing the challenges posed by noisy or sparse gradients: Stochastic Gradient Descent (SGD), Root Mean Squared Propagation (RMSProp), and Adam.

SGD, with its straightforward mechanism of updating weights by subtracting a portion of the gradient scaled by a fixed learning rate, offers simplicity and ease of implementation. However, its potential drawbacks include a slower convergence rate and a heightened sensitivity to the learning rate setting.

RMSProp builds upon SGD by introducing an adaptive learning rate for each weight, calculated using a moving average of the gradient's magnitude. This modification aims to accelerate convergence and reduce the learning rate's sensitivity, although it may lead to instability and divergence if the momentum term is excessively high.

Adam stands out by integrating the benefits of RMSProp's adaptive learning rate with Momentum's gradient descent acceleration, which relies on an exponentially decaying average of past gradients [39]. The advantage of Adam is that it can handle noisy or sparse gradients well and converge faster than SGD or RMSProp. Adam's formula for updating weights at each time step t is as follows:

$$\begin{aligned} \theta_{t+1} &= \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t} \\ \hat{v}_t &= \frac{v_t}{1 - \beta_2^t} \\ m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \end{aligned}$$

where, α represents the learning rate, $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected estimates of the first and second moments of the gradient g_t , β_1 and β_2 are decay rates that influence the memory effect, and ϵ is a small constant that prevents division by zero.

The comprehensive grid search revealed that the most effective hyperparameter combination for our model is 512 LSTM units, the Adam optimizer, and a learning rate of $1e-4$. This configuration demonstrated superior performance across all datasets, showcasing its ability to adeptly capture long-range dependencies and optimize the gradient descent process for enhanced efficiency. This finding underscores the model's robustness and its potential for handling complex sentiment analysis tasks.

Table 1

Performance comparison of different methods on the IMDb dataset.

Method	Accuracy	Precision	Recall	F1-score
CNN-BiLSTM-Attention [25]	91.41	89.75	90.57	90.18
BERT-CNN-LSTM-SVM [27]	93.4	93.4	93.4	93.4
BiLSTM [26]	88.81	–	–	–
caps-BiLSTM [26]	91.96	–	–	–
CNN-LSTM [24]	83.13	83.54	83.50	83.02
DistilBERT [20]	92.82	–	–	–
RoBERTa-LSTM [29]	92.96	93	93	93
BERT-base [8]	93.46	–	–	–
BERT-base+UDA [40]	94.55	–	–	–
XLNet-base [21]	95.05	–	–	–
RoBERTa-base [11]	95.06	–	–	–
DeBERTa-base [23]	95.12	–	–	–
Proposed Hybrid	94.83	94.8	94.8	94.8
Proposed Hybrid +Yelp-F	95.27	95.3	95.3	95.3

6. Experimental results and discussions

In this study, we conduct a rigorous evaluation of the performance of our RoBERTa-ResNeXt-BiLSTM model against a diverse set of state-of-the-art methods. This comparative analysis includes both classical and recent models, providing a comprehensive overview of the current landscape in sentiment analysis. The models chosen for comparison represent a range of methodologies in the field, from traditional machine learning approaches to the latest transformer-based models. This diversity allows us to benchmark our model against a broad spectrum of techniques, thereby demonstrating its effectiveness across different sentiment analysis paradigms.

We conduct our analysis on four sentiment analysis datasets: IMDb, TweetEval (sentiment), Yelp Polarity, and Yelp Review Full. These datasets were chosen as they have been widely used in the benchmark methods, ensuring a fair and consistent evaluation.

Our primary evaluation metric is mean accuracy, derived from three random seeds. This metric reveals that our model outperforms most of the compared methods and even surpasses the RoBERTa base model on two of the datasets. However, accuracy alone does not provide a complete picture of our model's performance. Therefore, we have employed additional evaluation criteria to assess the quality of our model more comprehensively:

- **Precision:** The proportion of correctly predicted positive observations out of all predicted positive observations.
- **Recall:** The proportion of correctly predicted positive observations out of all actual positive observations.
- **F1-score:** The harmonic mean of precision and recall, providing a balance between the two.
- **Accuracy:** The proportion of correctly predicted observations out of the total number of observations.

For a balanced comparison across different classes, we utilize a micro-average method, ensuring each class is weighted equally, irrespective of its size.

Our experiments are conducted on a robust system equipped with 32 GB RAM and a K40M GPU, which boasts 12 GB of memory and the capability to run up to 2880 CUDA cores at a clock speed of 745 MHz. We leverage PyTorch for our deep learning framework and the Hugging Face Transformers library for accessing pre-trained language models.

Our analysis of the IMDb dataset, as detailed in Table 1, reveals that our hybrid model achieves an accuracy of 94.83%, precision of 94.8%, recall of 94.8%, and an F1-score of 94.8%. While these metrics are slightly lower than those of leading methods such as RoBERTa base,

Table 2

Performance comparison of different methods on the TweetEval(Sentiment) dataset.

Method	Accuracy (%)
SVM [38]	62.9
Fast Text [38]	62.9
BiLSTM [38]	58.3
caps-BiLSTM [26]	66.42
GPT-3 [41]	58.4
GPT-J [41]	57.4
ChatGPT [42]	63.31
XLNet-base	68.6
RoBERTa-base [38]	71.3
Proposed Hybrid	71.56

XLNet base, and DeBERTa base, they significantly surpass other models like DistilBERT, BERT-base, BERT-base+UDA, and RoBERTa-LSTM. The efficiency of our model is underscored by its reduced parameter count of 27 million, compared to RoBERTa base's 125 million, and a shorter fine-tuning duration on the IMDb dataset (20,973 s as opposed to 27,729 s), thereby reducing computational complexity and memory demands.

Additionally, employing transfer learning, we further fine-tune our model, previously trained on the Yelp Review Full dataset, on the IMDb dataset. This results in an enhanced accuracy of 95.27%, precision of 95.3%, recall of 95.3%, and F1-score of 95.3%. These results not only top all other methods but also slightly exceed the DeBERTa base model, which, despite its high accuracy, requires the longest tuning time (45,323 s). This exemplifies the adaptability and efficiency of our hybrid model in applying knowledge acquired from extensive, detailed sentiment analysis datasets to varied domains and tasks.

In addition, we conducted a Friedman test on the six models that were applied with three random initializations. The resulting p -value was 0.030, indicating a statistically significant difference in the performance of these models on the IMDb dataset. To further investigate these differences, we conducted pairwise comparisons using the Wilcoxon signed-rank test and the Nemenyi test. For instance, the comparison between the RoBERTa base model and our proposed model yielded p -values of 0.25 and 0.36 from the Wilcoxon and Nemenyi tests, respectively. These results suggest that there is not a statistically significant difference in the performance of these two models on the IMDb dataset at the 5% significance level. In essence, our proposed model performs comparably to the RoBERTa base model, underscoring the robustness and reliability of our model.

In the TweetEval (sentiment) dataset, as shown in Table 2, our model registers the highest accuracy of 71.56%, marginally outperforming the RoBERTa base model (71.3%) and significantly outstripping other methods like caps-BiLSTM, SVM, and FastText. The model also reports uniform precision, recall, and F1-score values of 71.6%. These metrics suggest that our model adeptly navigates the complexities of sentiment analysis within social media content, effectively managing lexical diversity, informal language, and nuances such as sarcasm.

Our model, with its 27 million parameters, demonstrates superior performance despite having significantly fewer parameters than GPT-based models and the RoBERTa base model, which has 125 million parameters. For instance, GPT-J, a smaller variant of GPT models, has 8 million parameters, which is significantly fewer than GPT-3's 175 billion parameters [41]. Despite GPT-J's more feasible size for inference on consumer-level hardware, our model still outperforms it. Moreover, our model requires less fine-tuning time on the TweetEval (sentiment) dataset (8217 s), reinforcing its efficiency, effectiveness, and reduced computational footprint. This performance is particularly impressive when compared to the RoBERTa base model, which requires a longer fine-tuning time of 15,384 s. This highlights the efficiency and adaptability of our model.

Table 3

Performance comparison of different methods on the Yelp-P and Yelp-F dataset.

Method	Accuracy-Yelp-P	Accuracy-Yelp-F
VDCNN [17]	95.72	64.72
LSTM [43]	94.74	58.17
CNN [16]	94.7	62.0
FastText [33]	95.8	64.4
RNN-CNN [44]	94.5	61.8
Densely-Connected-CNN+Multi-scale Feature Attention [45]	96.5	66.0
Attention [46]	94.9	63.4
Att-BiLSTM [47]	95.02	63.23
CNN-GRU-Attention [43]	96.0	65.31
CCCaps-Net [48]	96.48	65.85
Roberta-base [11]	97.7	69.52
Proposed Hybrid	97.7	69.2

Table 4

Evaluating TS-SS and dot-product methods on IMDB and TweetEval(Sentiment) datasets.

Feature-dimensionality	Method	IMDb	TweetEval
512	TS-SS	94.528	71.08
512	dot-product	94.54	71.19
1024	TS-SS	94.63	71.33
1024	dot-product	94.83	71.56
1536	TS-SS	94.3	70.45
1536	dot-product	94.61	71.0

Furthermore, we conducted a Friedman test on the four models that were applied with three random initializations. The resulting p -value was 0.029, indicating a statistically significant difference in the performance of these models on the TweetEval dataset. Further pairwise comparisons using the Wilcoxon signed-rank test and the Nemenyi test were conducted. For example, the comparison between the RoBERTa base model and our proposed model resulted in p -values of 0.25 and 0.75, respectively. These results suggest that the performance difference between these two models on the TweetEval dataset is not statistically significant at the 5% significance level.

Table 3 presents a comparative performance analysis of the performance of various models on the Yelp Polarity and Yelp Review Full datasets. Our model demonstrates exceptional accuracy, achieving 97.7% on the Yelp Polarity dataset and 69.2% on the Yelp Review Full dataset. These figures are comparable to, or slightly behind, the performance of the RoBERTa base model but are notably superior to other methods such as CCCaps-Net, CNN-LSTM, and BiLSTM. The precision, recall, and F1-score of our model on the Yelp Polarity dataset stand at 97.7%, and 69.2% on the Yelp Review Full dataset, indicating its proficiency in conducting nuanced sentiment analysis and categorizing reviews across varying levels of satisfaction. Furthermore, our model is distinguished by a significantly lower parameter count (27 million) compared to the RoBERTa base model (125 million), and it requires less fine-tuning time on both datasets (313,196 s vs. 414,086 s), enhancing computational efficiency and reducing memory usage.

Table 4 evaluates the TS-SS and dot-product methods on the IMDb and TweetEval (sentiment) datasets. The results indicate that while TS-SS competes effectively with the dot-product method at lower feature dimensionalities, its performance diminishes as the dimensionality increases. This suggests that the effectiveness of TS-SS in capturing document similarity is contingent on maintaining a lower feature dimensionality.

Table 5 assesses the performance of ResNeXt and ResNet on the IMDb and TweetEval (sentiment) datasets. ResNeXt demonstrates a slight advantage over ResNet, although the difference is minimal. This implies that ResNeXt's ability to extract text features is comparable to

Table 5

Evaluating ResNeXt and ResNet methods on IMDb and TweetEval(Sentiment) datasets.

Method	IMDb	TweetEval
ResNeXt	94.83	71.56
ResNet	94.72	71.33

Table 6

Comparison of the number of parameters across models and datasets.

Method	IMDb	TweetEval	Yelp-P	Yelp-F
Roberta-base	124647170	124647939	124647170	124649477
Proposed Hybrid	27 191 874	27 191 874	27 191 875	27193413

Table 7

Comparison of fine-tuning time for different models on various datasets.

Method	IMDb	TweetEval	Yelp-P
Roberta-base	27729 s	15384 s	414086 s
Proposed Hybrid	20973 s	8217 s	313196 s

that of ResNet, with added benefits stemming from its simpler, more modular structure and enhanced representational capacity.

Our RoBERTa-ResNeXt-BiLSTM model encapsulates the essence of innovation in sentiment analysis, adeptly handling the dynamic and complex nuances of natural language across various text datasets. This hybrid model, with its strategic integration of robust feature extraction and precise sequential modeling, not only upholds the linguistic depth of pre-trained models but also showcases remarkable adaptability and computational efficiency (see Tables 6 and 7).

The empirical results affirm the model's edge over existing methods, delivering a compact yet comprehensive representation of textual features while significantly reducing computational demands. The inclusion of the TS-SS similarity measure, despite its empirical performance, reflects our commitment to exploring innovative methodologies and pushing the boundaries of sentiment analysis.

In summary, our model represents a significant advancement, balancing high accuracy with computational pragmatism. Its versatility and efficacy in training across varied datasets underscore its potential for practical applications in natural language processing. The model's meticulously engineered structure is poised to address the evolving challenges of sentiment analysis, solidifying its position as a cutting-edge solution in the field.

7. Conclusion

This study introduced a novel sentiment analysis model that synergistically combines RoBERTa, ResNeXt, BiLSTM, and self-attention mechanisms. By harnessing the strengths of these components, our model adeptly extracts salient features from text, elicits sequential information, and evaluates document similarity. Our comprehensive evaluation across four sentiment analysis datasets, spanning various domains and tasks, has demonstrated that our model not only achieves high accuracy but also significantly reduces training time compared to solely fine-tuning RoBERTa.

The experimental results have affirmed the model's efficiency, outperforming state-of-the-art methods with fewer parameters and less training time. In our exploration of document similarity measures, we found that TS-SS serves as a viable alternative to the dot-product when feature dimensionality is low, although its effectiveness diminishes with higher dimensionality. Additionally, the ResNeXt architecture has proven to be a potent tool for text feature extraction, offering advantages over ResNet such as a more modular structure and greater representational capacity. We believe that our model will extend its utility to other natural language processing tasks involving text classification, such as topic modeling, document clustering, and text summarization.

Looking ahead, we aim to extend the adaptability of the RoBERTa model to a broader spectrum of domains and languages, particularly in areas where pre-training data is scarce. We plan to devise methods for dynamically adjusting feature dimensionality to better accommodate the unique characteristics of different datasets. Additionally, we will explore the implementation of multiple self-attention layers or various types of such mechanisms to capture complex document relationships and hierarchical structures more effectively. The potential of employing a Siamese network with CNNs for similarity calculations within self-attention frameworks will also be investigated, leveraging its learnable nature. Furthermore, we intend to examine the impact of different depth and cardinality settings within the ResNeXt architecture on text feature extraction, optimizing it for enhanced performance.

CRedit authorship contribution statement

Amir Jabbar Lak: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation. **Reza Boostani:** Validation, Supervision, Investigation. **Farhan A. Alenizi:** Writing – review & editing, Validation, Funding acquisition. **Amin Salih Mohammed:** Resources. **Sayed Mostafa Fakhrahmad:** Validation, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have used public datasets for my research, namely TweetEval (sentiment), Yelp Polarity and Yelp Full, and IMDB. I have also mentioned the main articles that introduced these datasets in my paper.

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